

The Advantages Of An LCD Monitor

Luminance (Brightness): An LCD can be made brighter by increasing the brightness of the backlight. In the case of a CRT, an increase in brightness also increases the beam spot size; this lowers the effective resolution, yielding a fuzzy image.

Flickerless: An LCD display does not flicker. In a CRT, an electron beam scans the phosphor screen in horizontal lines from top to bottom and this illuminates the screen. This must be done fast enough for the eye to not notice a flicker, which is not always achievable. The LCD has a constant source of light over the whole screen which means that once a pixel is on, it stays on until turned off, and vice-versa.

Crisp And Sharp Image: For a CRT screen, the electron beam is circular when aimed directly forward, but it becomes elliptical when aimed at any other direction. This may cause image clarity or focus issues at the screen edge.

An LCD has millions of pixels hard-etched onto the panel, each effectively independent from its neighbour. With no scanning electron beam, distortion is not an issue. Therefore, the image always appears crisp over the entire screen.

Perfect Geometry: LCD monitors provide geometrically-perfect and distortion-free images, which is a huge advantage for graphics artists and designers.

Longer Life: The life of an LCD depends on the life of the backlight, which is composed of one or more tiny fluorescent tubes. The typical life of a backlight is stated to be around 50,000 hours to the half-brightness point, which is the point at which brightness is one-half the original brightness. This is the industry standard measure for product life.

In case of a CRT, the age depends on the formation of oxide layer on the cathode of the electron gun, which decreases beam current. Also, the phosphor ages and becomes less efficient over a period. The typical CRT half-brightness point has been found to be between 10,000 and 20,000 hours.

Power Consumption: The power required to run an LCD is about one-half to one-third of that required for a CRT with the same screen area. In addition to this, LCDs generate considerably less heat than CRT monitors because of the technological difference. This indirectly leads to a lower load on air conditioning, thus reducing running costs. Similarly, the lower power requirements of an LCD

monitor make it easier to be used with an uninterruptible power supply, because the lower power required provides precious extra minutes to save important data and shut down the PC in the event of a power outage.

Low Emissions: A CRT monitor can generate electric, magnetic and even X-ray emissions due to the high-voltage power supply necessary to drive the CRT and the electron beam striking the phosphor screen. An LCD has no such issues because of the different technology.

Better Ergonomics: LCD monitors are smaller, more compact and lighter when compared to CRTs. Thus, they take up less precious desktop space. They can fit in locations where bulky CRTs just cannot. Those with VESA mounting can even be mounted on the wall. Some LCDs also have pivots to rotate horizontally, vertically and even from landscape to portrait mode, which makes them more versatile.

Total Cost Of Ownership: The prices of LCD monitors have been decreasing in recent years. When calculated, one can imagine the power-saving features of LCDs reducing the total cost of ownership over a period of a few years.

because of its overall better image quality. It was priced at Rs 14,000, which is just below the average price of a 17-inch monitor. The Philips 170S was awarded the Digit Best Buy Silver.

19-INCH LCD MONITORS

We found the NEC monitors to be a world apart from the others, due to their better features and quality. But they were also priced exorbitantly high, and remain beyond the reach of most.

The NU L921G had high brightness and contrast ratios, and a low response time, while weighing just

5.3 kg, scored good points under Features. It also fared well as far as performance was concerned. The low price tag of Rs 18,000 was the deciding factor in adjudging the NU L921G the winner of the Digit Best Buy Gold in the 19-inch LCD monitor category.

The Asus PW191 was in a class of its own with its wide-screen aspect ratio and unique looks. It scored the highest points in Features. We also found it to be a good performer. But with its heavy price tag of Rs 29,000, it had to settle for the Digit Best Buy Silver.

In Conclusion

We'd like to confess that we had initially planned to carry out two separate shootouts of both LCD monitors and CRTs, but we were surprised at the very low turnout of CRTs. Manufacturers told us they are pushing for sales of LCDs over CRTs, hence the low turnout.

The wisdom behind this decision is debatable. If you recall our last LCD shootout, you would realise that the prices have not plummeted as much as we had hoped they would, and India is a price-sensitive market. People will still want to buy CRTs.

If you have noticed, there were many more 17- and 19-inch monitors than 15-inch ones. To us, it seems like manufacturers are bent on dictating purchasing terms.

We can only hope for a significant drop in LCD prices the next time round. Better technology should also bring in better response times. The transition from CRTs to LCDs would then be more likely. ■

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